

DSLAb-syntheticdata

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Abstract

Data security is crucial in all fields. However, data collection can be challenging due to issues such as data sensitivity and privacy. Furthermore, as some countries face constraints in data collection during their economic development, the generation of synthetic data is becoming increasingly important. Recently, synthetic time-series data models have been expanded upon based on VAE, diffusion, and generative adversarial networks (GANs). Models such as TimeGAN, conditionalGAN, and Chronos-Diffusion have demonstrated excellent performance in addressing these shortcomings. However, due to the long survey cycles of economic development indicator time series, limited data sets, and high levels of missing data, the use of GANs has been challenging. Therefore, in this study, we applied the TIMEVAE model, which demonstrated efficient generation capabilities, to generate synthetic data for economic development indicators. We then examined which model, CSDI or SAITS interpolation, produced better results to improve the performance of the synthetic data. This study demonstrates the significance of addressing synthetic data for economic development indicators in the field of synthetic data and is expected to contribute to the development of improved synthetic data for economic development indicators and policy simulations.

1 Introduction

Securing meaningful data is essential for better decision-making. However, unlike ideal circumstances, securing sufficient data can be challenging due to various reasons, such as capital or time constraints. In such cases, data is typically generated using traditional statistical methods that consider the underlying data distribution, or rule-based machine learning (ML) techniques such as averages, KNN, and SMOTE. Recently, synthetic data is increasingly being generated through deep learning. Synthetic data is virtual data. This data is utilized in a variety of fields. According to Yingzhou Lu et al. (2025)[1], synthetic data

is used in research across various fields, including voice, video, medicine, business, and finance. This data is useful when data is scarce or absent. It also has the advantage of being able to replace sensitive information. Despite these advantages, research on synthetic data in the field of economic development remains limited. While data acquisition is crucial for making rational decisions in economic development, data availability is particularly limited in developing countries. This is because research is sometimes impossible due to various constraints, such as war, financial difficulties, and system failures. Moreover, data related to economic development is not the large-scale data measured in minutes or seconds used in industry, but rather macroscopic time series data spanning long periods, such as one year or quarter. Consequently, data limitations pose challenges for model training. Therefore, this study utilized SDG and Finance data provided by World in Data, applied CSDI and SAITS interpolation models, and compared the results with TimeVAE, an existing synthetic data model. This was used to determine which interpolation model was superior in producing improved data for economic indicator data generation, using three evaluation methods. This approach is expected to address data shortages in the field of economic development, identify effective models, and provide improved data for policy simulations, thereby contributing to the development of efficient systems. Ultimately, we hope that the development of synthetic data will lead to national development and an improved quality of life for individuals.

2 Literature Review

Data scarcity is a well-known fact in the economic field. According to Allison Koencke et al. (2020), [2] economic data is difficult to utilize due to privacy, sensitive information, or access limitations. This makes the replication of research findings difficult.

The African Centre for Economic Transformation (ACET) [3] stated that Africa is currently using global data to make decisions that are not fit for purpose. Research is underway to generate synthetic data that resembles reality, rather than collecting new data to reduce the high cost burden and utilize data directly.

This suggests that, despite economic development being a critical national priority, there is a lack of relevant research and concrete models.

Research on synthetic data has been conducted in the financial sector. Swachha Sisir Das et al. (2025) [4] used conditional GANs to generate data on Indian street vendors. They observed results such as a small Frechet distance between the real and synthetic data. While using conditionals significantly improved efficiency, this data was inherently private and therefore lacked reproducibility.

Cemal Oztrk et al. (2024) [5] used a TimeGAN model to predict stock values based on Google stock price data. This demonstrates that TimeGAN can learn from complex data such as stocks. However, it should be noted that this study used stock data, which is massive in scale and may not yield similar results when the data is small.

Research on various models related to synthetic data is ongoing.

Xiamen Li et al. (2022)[6] propose a TTS-GAN model using Transformer to address the challenge of collecting physiological time-series signals in the medical field, where data collection is challenging. This model reduces the temporal dependence of the TimeGAN model, which was first studied in the late 2010s.

Sometimes, a VAE model is used instead of a GAN. GANs require a large amount of training data and have a long learning time. To improve time efficiency, the existing encoder structure utilizes trend and seasonal blocks as decoders, omitting the RNN and GAN used in TimeGAN and facilitating interpretation.

Paul Jeha et al. (2022) [7] developed a Progressive Self-Attention GAN model that utilizes self-attention to capture long-term time series dependencies and generate long-term time series data, overcoming existing limitations.

Recently, diffusion models have been introduced beyond GANs. Marcel Kollovich et al. (2023)[8] reported that they outperformed VAE and GAN models by leveraging the characteristics of diffusion models to learn the entire time series distribution.

Meanwhile, TSGM[9], announced in 2024, contributed to increasing consistency among the growing number of synthetic time series data generation models with diverse objectives and evaluation methods by utilizing a comprehensive time series synthetic data generation framework with a powerful model and 140 datasets.

In the field of synthetic data, the development of models for data generation is largely centered around VAE, GAN, and Diffusion. Each model offers distinct advantages, such as faster learning, tabular data generation, or time series data generation.

Expanding on the concept of synthetic data generation, we can consider the concept of "interpolation." Unlike synthetic data, which creates a new dataset based on existing data, interpolation maintains the existing data but creates empty, missing values. It parallels the context of synthetic data in that it creates new data.

Recent interpolation research exemplifies the CSDI and SAITS models.

CSDI is an interpolation model based on a diffusion model that learns data by filling in missing values with noise and then generating missing values. It is known to excel in medical and environmental time-series data.

SAITS is an interpolation model that uses self-attention to learn correlations between features. It is known for its high accuracy and efficiency compared to its speed.

However, most models are still developed primarily with clean, complete data, such as electricity and transportation data, and even in the economic field, only data without missing values, such as stock data, are used. There have been no cases of utilizing data collected at the national level from international organizations. This study aims to overcome these limitations by appropriately utilizing interpolation and synthetic data generation models.

3 Methodology

3.1 Framework

This study interpolated preprocessed data using the CSDI and SAITS models, removing missing values and storing the data as complete data. The data was then applied to the TimeVAE synthetic data model. This study aimed to determine which interpolation technique was most effective in generating improved quality synthetic data.

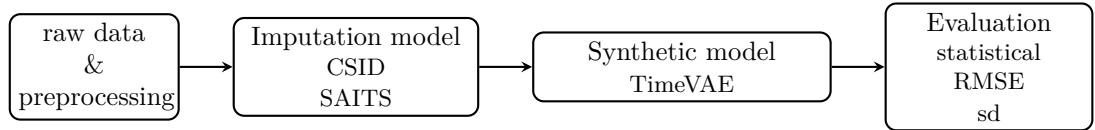


Figure 1: Flow

3.2 Data

3.2.1 Data Collection & Preprocessing

This study used data related to international economic development indicators provided by the World Bank Open Data. The initial data included indicators related to the SDGs and global financial development, including the following. Some indicators were removed during preprocessing. These indicators allow for the study of not only SDG areas highlighted by international organizations, but also direct economic indicators.

- Access to electricity (% of population)
- Annualized average growth rate in per capita real survey mean consumption or income, total population (%)
- Births attended by skilled health staff (% of total)
- Coverage of social insurance programs (% of population)
- Debt service (PPG and IMF only, % of exports of goods, services and primary income)
- Employment in industry (% of total employment) (modeled ILO estimate)
- Account at a formal financial institution (% age 15+)
- Bank accounts per 1,000 adults
- Bank deposits to GDP (%)
- Central bank assets to GDP (%)

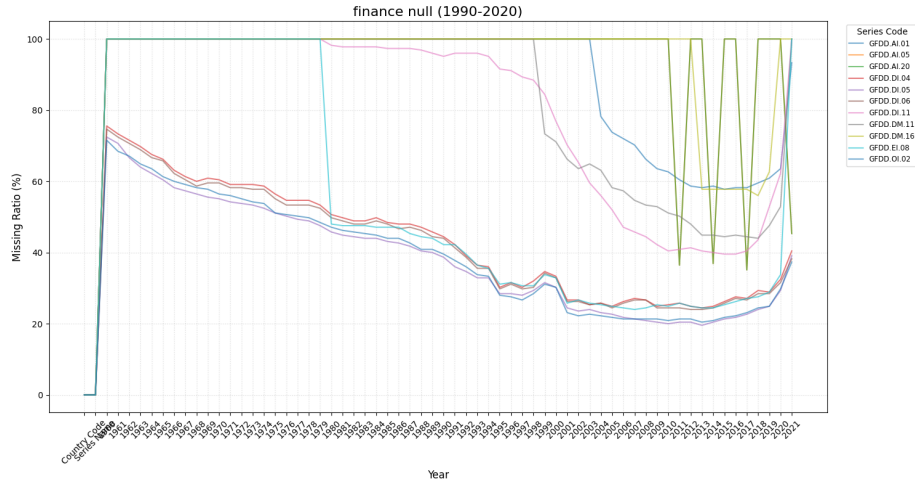


Figure 2: architecture diagram

- Credit card (% age 15+)
- Credit to government and state-owned enterprises to GDP (%)
- Credit flows by fintech and bigtech companies to GDP (%)
- Deposit money bank assets to deposit money bank assets and central bank assets (%)
- Insurance company assets to GDP (%)
- Gross portfolio debt assets to GDP (%)
- Liquid liabilities to GDP (%)

Upon reviewing the data, we found that the SDG data spanned from 1990 to 2020, while the Finance data spanned from 1960 to 2021. Therefore, we preprocessed the data to cover the same period, from 1990 to 2020. While there were concerns about data loss, most of the missing data in the deleted period exceeded 50%, suggesting a potential for distortion even with interpolation. Therefore, we deleted the data.

Furthermore, because the country data for the two indicators were inconsistent, the two indicators were merged based on the 214 countries they shared. After merging, five indicators with initial missing values exceeding 80% were removed to account for distortions in interpolated data. The deleted indicators are as follows:

- Annualized average growth rate in per capita real survey mean consumption or income, total population (%)

- Coverage of social insurance programs (% of population)
- Credit flows by fintech and bigtech companies to GDP (%)
- Credit card (% age 15+)
- Account at a formal financial institution (% age 15+).

When we checked the missing value rate by mapping it with the country-specific income group data provided by the World Bank, we found that the missing value rate in high-income countries was surprisingly high. This is because the indicators used in this study are mainly indicators for the development of low-income countries, and the need for surveys in highly developed countries among high-income countries is low, so we had to consider cases of missing values.

Table 1: Missing vlaue percentage by Income Group

Income Group	Missing value(&)
High Income	47.12
Low income	38.27
Upper middle income	33.61
Lower middle income	31.22

Table 2: Average Number of values on poverty data

Income Group	Number of Year
Low	8.1
Middle	13.3
High	20.8

Therefore, to assess the validity of this study, prior to using the data, we separately reviewed daily median income and consumption data provided by the World Bank. The indicators below count cases where data was available in the given year. While low-income countries had an average of eight years of data, high-income countries had an average of approximately 20 years. This confirmed the need for data generation not only for low-income and underdeveloped countries, but also for high-income and highly developed countries.

After preprocessing the data, scaling and tensor processing were performed for model learning, and after interpolation, reverse scaling was performed to save the data in two versions, CSDI and SAITS, and a synthetic data model was applied.

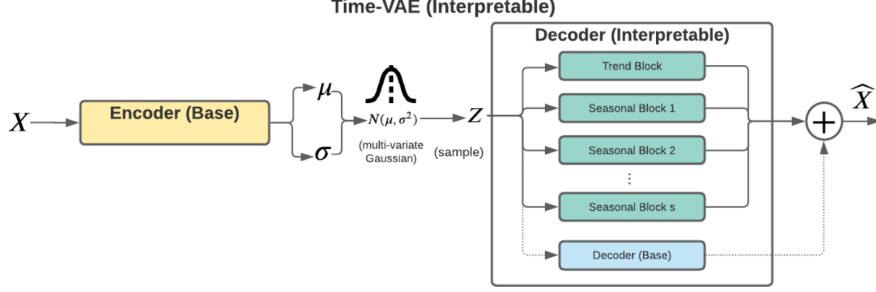


Figure 3: TimeVAE structure

3.3 Model

3.3.1 Imputation Models

CSDI: During training, a portion of the existing data is randomly processed as noise and the corresponding noise portion is predicted, thereby learning the structural patterns of time-series data. In other words, training continues toward minimizing the difference between the actual noise (ϵ) and the noise predicted by the neural network (ϵ_θ), following the equation below.

$$\min_{\theta} \mathcal{L}(\theta) := \mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), t} \left\| \epsilon - \epsilon_\theta(\mathbf{x}_t^{\text{ta}}, t \mid \mathbf{x}_0^{\text{co}}) \right\|_2^2 \quad (1)$$

SAITS: A time series interpolation data model based on self-attention. It learns by considering the correlation between patterns and features of time series data while masking future time points ($t+1$ and later) to refer only to the past time point and the present time point.

3.3.2 Synthetic Models

TimeVAE[10]: This is a time-series synthetic data generation model that extends the existing VAE model to identify partial patterns from the encoder to the convolution layer and reflect the temporal trend of the decoder.

3.4 Evaluation Metrics

Traditional statistical evaluation metrics include the mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), and model variance explained (R2).

MAE is the average of the absolute values of the differences between the actual and predicted values. It is characterized by being less sensitive to outliers.

The coefficient of determination (R2) is an indicator that measures the explanatory power of a model. It ranges from 0 to 1, with a higher value indicating a better model representation of the data. The mean squared error (RMSE) has

the advantage of being more sensitive to errors than the MAE. However, due to its high variability, it is also analyzed with the root mean squared error (RMSE) metric, which is interpreted in terms of the target variable unit.

The equations are as follows:

$$\begin{aligned} \text{MAE} &= \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \\ R^2 &= 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \\ \text{RMSE} &= \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \end{aligned}$$

4 Result

To confirm the superiority of the CSDI and SAITS models, this study examined traditional statistical evaluation methods, such as MSE and R2, and visualized overall distribution trends. Due to differences in units between data series indicators, the evaluation was based on scaled values. Furthermore, a 10% validation value was set aside during training, resulting in a final comparison of 192 countries.

4.0.1 Statistical Evaluation

Table 3: Result of Statistical Evaluation

Model Combination	MAE	R2	RMSE	MSE
TimeVAE + CSDI	0.080034	0.811514	0.151645	0.022996
TimeVAE + SAITS	0.099972	0.703990	0.183219	0.033569

As a result of checking the statistical evaluation index values, the combination of TimeVAE+CSDI numerically produced better results than the SAITS combination.

4.0.2 Visual Inspection

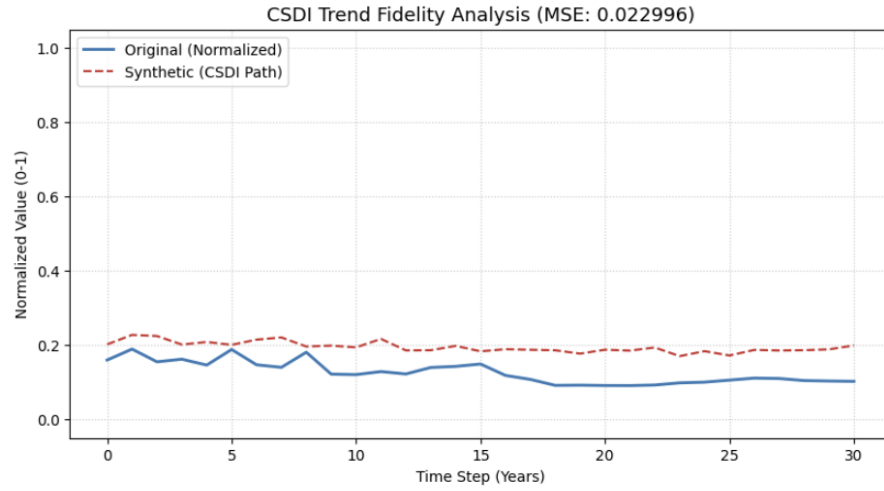


Figure 4: csdi

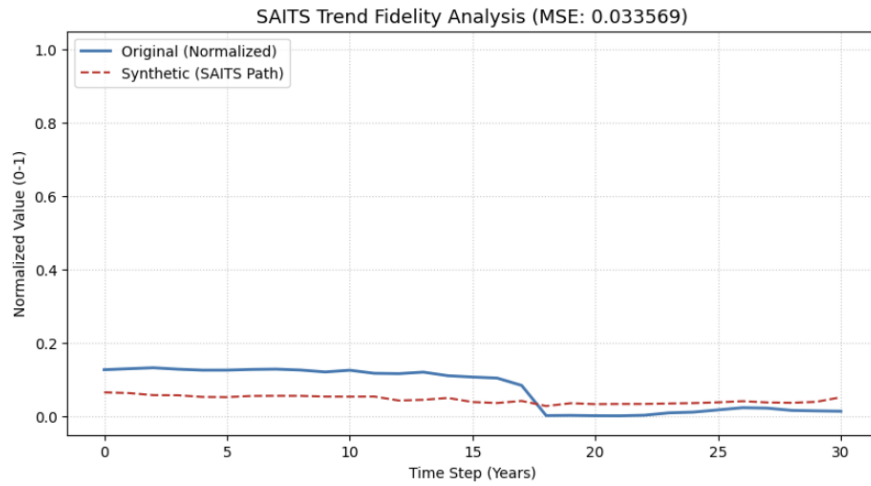


Figure 5: saits

These are the average predicted values by year for each interpolation model. Since these values are scaled, they appear relatively flat overall, so I examined them in detail for each series.

Yearly Trend Fidelity Analysis: TimeVAE + CSDI

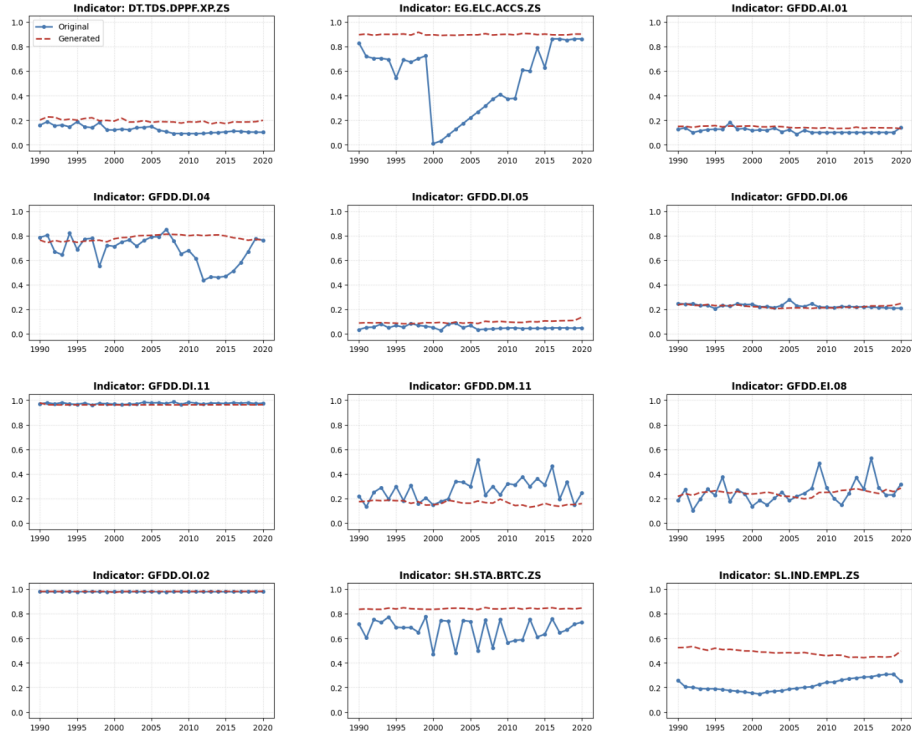


Figure 6: Afghanistan csdi

The pattern varies slightly by country. For example, the first country has the most unstable predictions for the EG.ELC.ACCS.ZS indicator, or Access to Electricity (% of population), in the series.

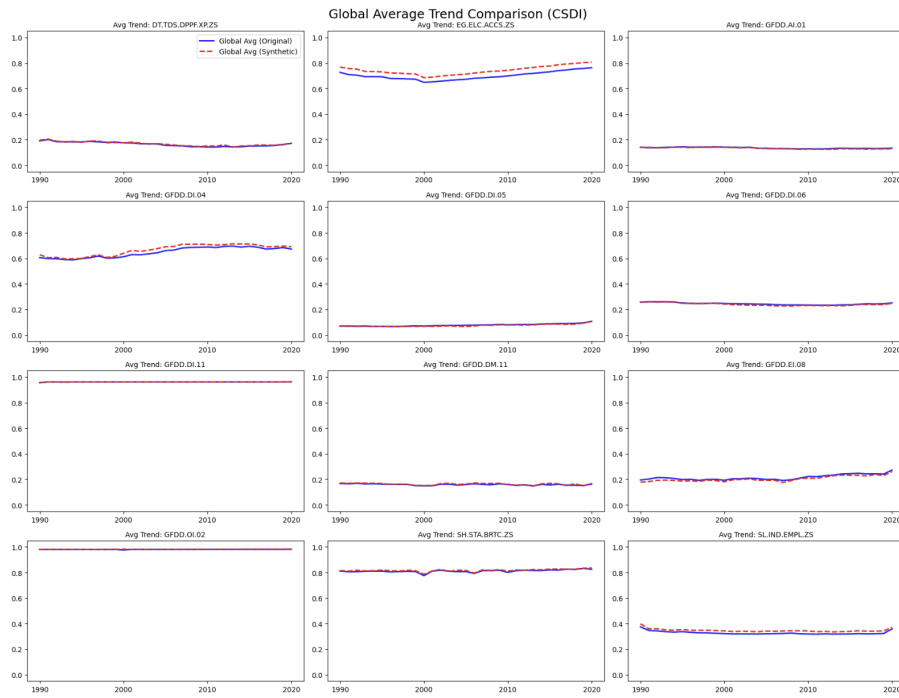


Figure 7: series csdi

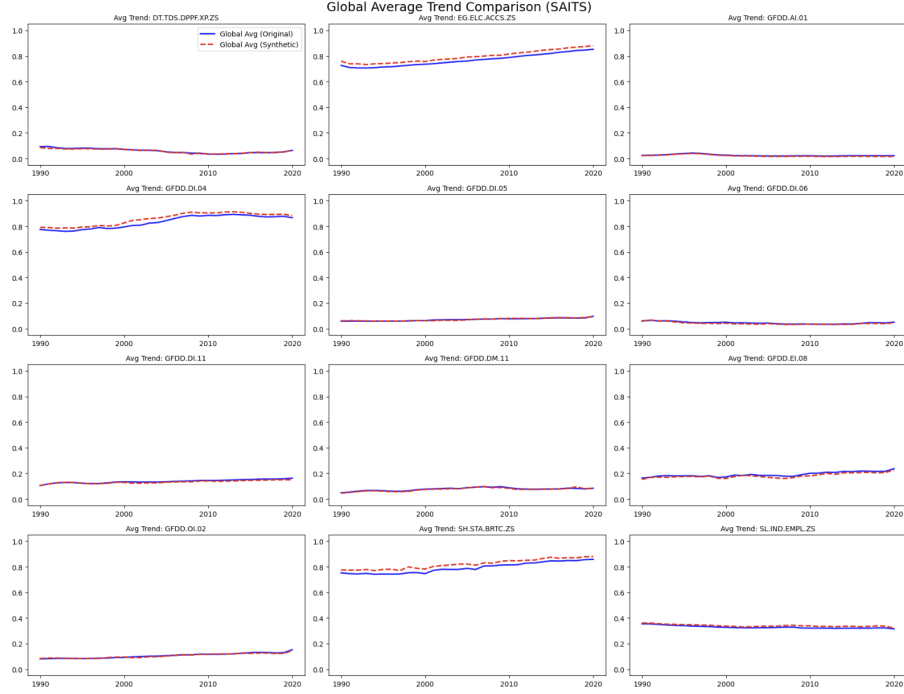


Figure 8: series saits

However, overall, the national average trends by series and year are similar, so it is judged that the trend of the synthetic data well reflects the trend of the existing data.

5 Conclusion, result

This study utilized economic data provided by the World Bank to secure economic indicators for underdeveloped countries with significant missing values, overcome data imbalances between countries, and provide a dataset for policy simulation and analysis previously impossible due to data shortages. Among the two models, the combination of CSDI and Time VAE outperformed the combination of SAITS+TimeVAE.

However, limitations and future challenges were identified during the research process. First, when attempting to further compare and verify the advantages of various synthetic data generation models, limitations such as inter-framework compatibility issues hindered extensive research. Second, while the CSDI interpolation model for handling missing values is known to be robust for medical data, considering that the data used in this study was economic data, further research on interpolation model development is needed. Third, research that differentiates the handling of missing values between data sets and

initial missing values is expected to lead to improved models for synthetic data generation. Future research could explore not only the improved VAE model, which overcomes compatibility issues, but also compare it with various other models. In addition, with regard to handling missing data, we will conduct research that considers the characteristics of each missing value and develops a new model that is robust to economic data, rather than the previously provided model.

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